Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/29/2007
PI Name:	Liu, Yuanming Ph.D.		
Project Title:	He-4 Studies in a Low-gravity Simulator		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	818-354-5998
Organization Name:	NASA Jet Propulsion Laboratory		
PI Address 1:	4800 Oak Grove Drive		
PI Address 2:	M/S 79-24		
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109-8099	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	10/01/2002	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Project extended to 9/30/2007 per D. Strayer, JPL (3/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	none		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The task is a ground-based, scientific research experiment, which will study the effects of a heat current on the helium-4 superfluid transition in a low-gravity simulator. This system represents a class of condensed matter physical systems that are driven out of equilibrium and exhibit critical dynamic behaviors near the phase transitions. This area of research has recently attracted more theoretical and experimental attentions that have yielded exciting new developments and new understandings. However, the agreements between experiments and theories are mostly qualitative; large discrepancies still exist. It is widely believed that gravity in earth-bound experiments may play a significant role in creating those discrepancies. Using the low-gravity simulator we have developed at JPL, we will be able to study the critical dynamic properties of helium and the superfluid transition more closely to the transition temperature in a reduced gravity environment. Also we		

	will be able to investigate the intrinsic effects of gravity on the physical properties under study and the artificial influences on the measurements themselves. We believe the results of the proposed experiment will benefit the low temperature helium physics community at large and the fundamental microgravity physics community in particular.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will demonstrate the research benefit of low-gravity simulator that utilizes magnetic levitation to counter gravity on a body of fluid.
Task Progress:	To be updated at the final report
Bibliography Type:	Description: (Last Updated: 04/09/2008)
Abstracts for Journals and Proceedings	Liu Y, Strayer D, Zhu D, Hollen S, Hays C, Israelsson U. "Variable gravity testbed to study habitation thermal fluid devices, gravity effects on biological systems, and combustion." Habitation 2006, Orlando, FL, February 5-8, 2006. Habitation 2006;10(3-4):219-20. , Feb-2006
Abstracts for Journals and Proceedings	Liu Y, Strayer D, Hollen S, Hays C, Zan J, Cohen EJ, Israelsson U. "Variable gravity simulation by Maglev." International Two-Phase Thermal Control Workshop, The Johns Hopkins Applied Physics Laboratory, MD, September 19-21, 2006. International Two-Phase Thermal Control Workshop, September, 2006. , Sep-2006
Abstracts for Journals and Proceedings	Liu Y, Strayer D, Hollen S, Hays C, Zan J, Cohen EJ, Israelsson U. "Variable gravity simulation by Maglev." 24th Space Simulation Conference, Annapolis, MD, November 6-9, 2006. 24th Space Simulation Conference, November 2006. , Nov-2006